

Brock Allan

bballan@uci.edu | 951-655-0133 | [BrockAllan/LinkedIn](#) | Irvine, CA 92617 | [BrockAllan/Github](#)

Education

University of California, Irvine

Bachelor of Science in **Computer Science**, GPA 3.6

- **Coursework:** Data Structures & Algorithms, Security & Privacy, Information Retrieval, System Design, Operating Systems, **Computational Complexity**, Differential Equations, **Machine Learning**
- **Distinctions:** Campuswide Honors Collegium, Published AI & Game Design Thesis, Dean's Honor List, CLAP certificate, Chancellors Excellence, FFA President, NHS President

Skills & Tools

- **Coding Languages:** Python (PyTorch, NumPy), Java, JavaScript, C, C++, C#, Prolog, Assembly, R
- **Industry Software:** Blender, Unity, Unreal Engine, Photoshop, Vim, Git, Spreadsheets, GDB
- **Industry Models:** Diffusion Models, SDEdit, UNet Architecture, Bilinear Interpolation

Technical Experience

Limitations of AI & LLMs in Game Design Methodology – Thesis

September 2025 – June 2026

- Led Advanced AI Research & Agile Workflows via acting as the Product Owner under the **Scrum framework**. Led a multi-stage empirical study evaluating the capabilities and structural constraints of state-of-the-art LLMs (including **GPT-4.3** and **GPT-5.5**), managing iterative sprints, analyzing critical LLM anomalies like **context drift** and **token-level optimization**, and adapted pipelines across sequential interactions.
- Designed and evaluated complex natural language programming strategies ranging from **Zero-Shot** and **Few-Shot** to **Simulated Tuning (Chain-of-Hindsight and Self-Correction paradigms)**, to optimize code quality, reduce intrinsic and extrinsic hallucinations, and mitigate downstream algorithmic defects in generative systems.
- Codified a multi-dimensional quantitative rubric mapping 6 distinct variables (Clarity, Flow, Balance, Length, Integration, Fun) onto a **1–7 scale** to empirically benchmark human-AI interaction. Isolated critical performance deltas demonstrating that guided context-aligned prompt engineering raised user experience metrics to a peak score of **6.33 / 7.0**.

Generating New Forms of Informal Expressions

January 2026 - March 2026

- Led a research project that explored the capability of **LLMs** to produce informal or novel expressions, including idiomatic and slang, using **Single-shot, Few-shot, and Fine-Tuning**.
- **Gemma-3-4b-it**, was fine tuned using **Low Rank Adaptation and Cross Entropy Loss**, then evaluated for Uniqueness, Creativity, Parts of speech, and Semantic Fluency.
- Gained hands-on expertise in working with models from **HuggingFace** and tuning those models for use in other scenarios.

Sentiment Analysis of IMDB Movie Reviews

September 2025 - January 2026

- **Classified** movie reviews as either positive or negative based solely on textual content.
- Implemented **Bag-of-words** using bigrams, **Unsupervised** learning through K-means ++ Clustering, and a **Random Forest** with **XGBoost** classifiers.
- The model reached **86.9%** testing accuracy after **hyperparameter optimization** and **5-Fold Cross Validation**.

Professional Experience

Learning Assistant, Intro to Programming, Shannon Alfaro, UCI

September 2025 - Current

- Instructed College students in how to code and design programs to achieve **foundational proficiency in core computing concepts**.
- Closely guided two groups of 6 students through technical coursework, hosted regular **office hours** to clarify complex topics and provided **detailed feedback** on assignments.
- Directed the LA team to **streamline communication**, provided **actionable feedback** to course instructors, and served as the **primary student advocate** to drive course improvements.